Development of a restaurant table management web application

Possible Subtitle

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Bachelor's thesis
December 2024
Degree Programme in International Logistics

Jyväskylän ammattikorkeakoulu
University of Applied Sciences

jamk

Description

Skakun, Aleksandr

Development of a restaurant table management web application

Jyväskylä: Jamk University of Applied Sciences, December 2024, 40 pages.

Degree Programme in International Logistics. Bachelor’s thesis.

Permission for open access publication: Yes

Language of publication: English

Abstract

Table reservation and management are essential parts of restaurant business. Efficient performance in both areas can improve customer experience and increase overall effectiveness of restaurant operations. However, many businesses are still relying on outdated and rudimentary methods to accomplish these tasks, which are prone to human errors and restrict operational efficiency, scalability and sustainability. Thus, table management applications were explored with a goal to identify essential functionalities and assess how emerging applications in this field can be developed and improved.

Through a combination of qualitative analysis of customer feedback and a detailed review of leading table management applications, the study highlights essential features of table management applications, which include guest database, table reservation tracking, restaurant layout visualization, and advanced analytics tools. These functionalities form the foundation of competitive table management platforms and should be prioritized during the development process.

The findings provide actionable insights for software developers working in this field, enabling them to design more effective and user-centered solutions. Additionally, the research serves as a practical guide for restaurant managers by equipping them with the knowledge needed to make informed decisions when selecting table management applications and the information about the benefits the software provides.

Keywords/tags (subjects)

Restaurant management, reservation tracking, table management application, software development

Miscellaneous (Confidential information)

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1 Introduction

Restaurants process high number of customers daily. A lot of establishments provide their customers with an option to book a table for certain date and time, so the customers do not have to deal with waiting time and could be seated the moment they arrive. However, this makes table management process even more complex as the staff must manage not only currently occupied tables but also keep track of reserved time slots to make sure that there is an available space for customers at appointed time and table reservations do not clash with each other. For this purpose, restaurant management and hosting staff adopt different ways of tracking reservation. Though, in many cases these systems are rudimentary and inefficient. For example, some places keep track of the reservations in excel or even just by using pen and paper. This approach has many drawbacks. It is very inefficient as staff need to make each entry fully by hand. Also, this system is prone to errors and might lead to negative customer experience. What is more, it makes the process of gathering data about the customers and restaurant operations challenging.

To solve this problem table management applications and CRM systems exist. This research strives to explore table management applications functionalities and what functions are required to make an effective service and provide the most value for restaurant management staff. The thesis focuses on accessing current landscape of table management applications and how such services can be improved. One of the companies providing services in this field is SeatsFlow. They develop table management application for restaurants and cafes. The application provides hosting staff with a table management interface to make booking process more reliable and efficient. SeatsFlow has relatively new and quickly developing application. Its small size makes it a good target to explore further development and improvement process of table management applications. Thus, the thesis focuses on accessing necessary qualities for the software by performing a research-based development to gain knowledge about table management applications while working on improving the state of SeatsFlow's table management application.

1.1 Research objectives and questions

The goal of the project is to establish customer requirements for restaurant management applications and explore opportunities to develop a more efficient table reservation process to gain competitive advantage in the field of table management systems. What is more, it is required to consider current state of the application and development methodology adapted for the project, so the solutions can be applied to real life situation. This research focuses on analyzing current trends in table management software and providing comprehensive analysis of the state of the application and improvement strategies. The study will answer the following questions:

- What development methodology can be suitable for a small-scale application?
- What functionalities are usually present in table management applications?
- How can the application in question be improved to be competitive on the market?
- What features should be prioritized while developing table management application?

This research assists people developing a table management application to gain an understanding of the development process. Also, it can help restaurant business owners to analyze the functions of table management applications to make a more informed decision when choosing one for their business. What is more, it provides clear view of development and improvement process for a small-scale digital service. In addition, this project helps to achieve goal 9 of sustainable development by enhancing operational efficiency of restaurants with technology. Furthermore, the research is relevant to goal 12 of sustainable development by assisting restaurants in reducing waste and improving resource utilization efficiency.

The study adopts qualitative research methods by analyzing feedback of the customers who tested the application. In addition, it accesses the state of the competitors in the same field to acquire necessary data as a basis for the research. The research combines case study and grounded theory methods to answer the posed questions. Merriam (2002) describes case study as an analysis of individual case to achieve the insights into a broader concept and grounded theory as an extraction of insights from data to form a theory which by definition is based on data. Thus, it allows to explore table management applications and form a theoretical basis for decision making. This helps to determine the best strategy for the development process as well as deepen the understanding of requirements for table management software.

2 Software development approaches

To determine the best direction for application development it is necessary to consider the development methodology adopted for the project. Based on the chosen development cycle it becomes possible to determine what strategy would be appropriate. There are multiple methodologies for software development projects. Some of them favor slower development cycles with higher focus on designing and planning stages and more time for implementation of features in big batches. On the other hand, there are methodologies that focus on adaptability and faster development cycles to deliver updates to the customer as fast as possible. Stepanov (2021) argues that three classic methodologies for software development are waterfall, spiral and iterative and other methodologies were developed based on these three. The iterative methodology sometimes is used synonymously with agile. Even though they are not exactly the same and other traditional models can follow iterative development process, it is not required to delve into the differences between them to determine the most appropriate development approach for the project. Thus, the research will focus on advantages and disadvantages of waterfall, spiral and agile methodologies and their application for development table management solutions. These methodologies cover wide range of approaches to software development including sequential and iterative process and can be applied to the projects of various sizes.

2.1 Waterfall methodology

Waterfall methodology is one of the oldest software development models. According to Ruparelia (2010) the methodology was developed in 1956 by Benington and modified in 1970 by Royce. The original model had a rigid one-way design, so Royce felt that it needed to be modified to add feedback loops, so that previous steps could be revisited. However, Royce also believed that one step revision approach may be insufficient, thus more complex feedback structure was developed.

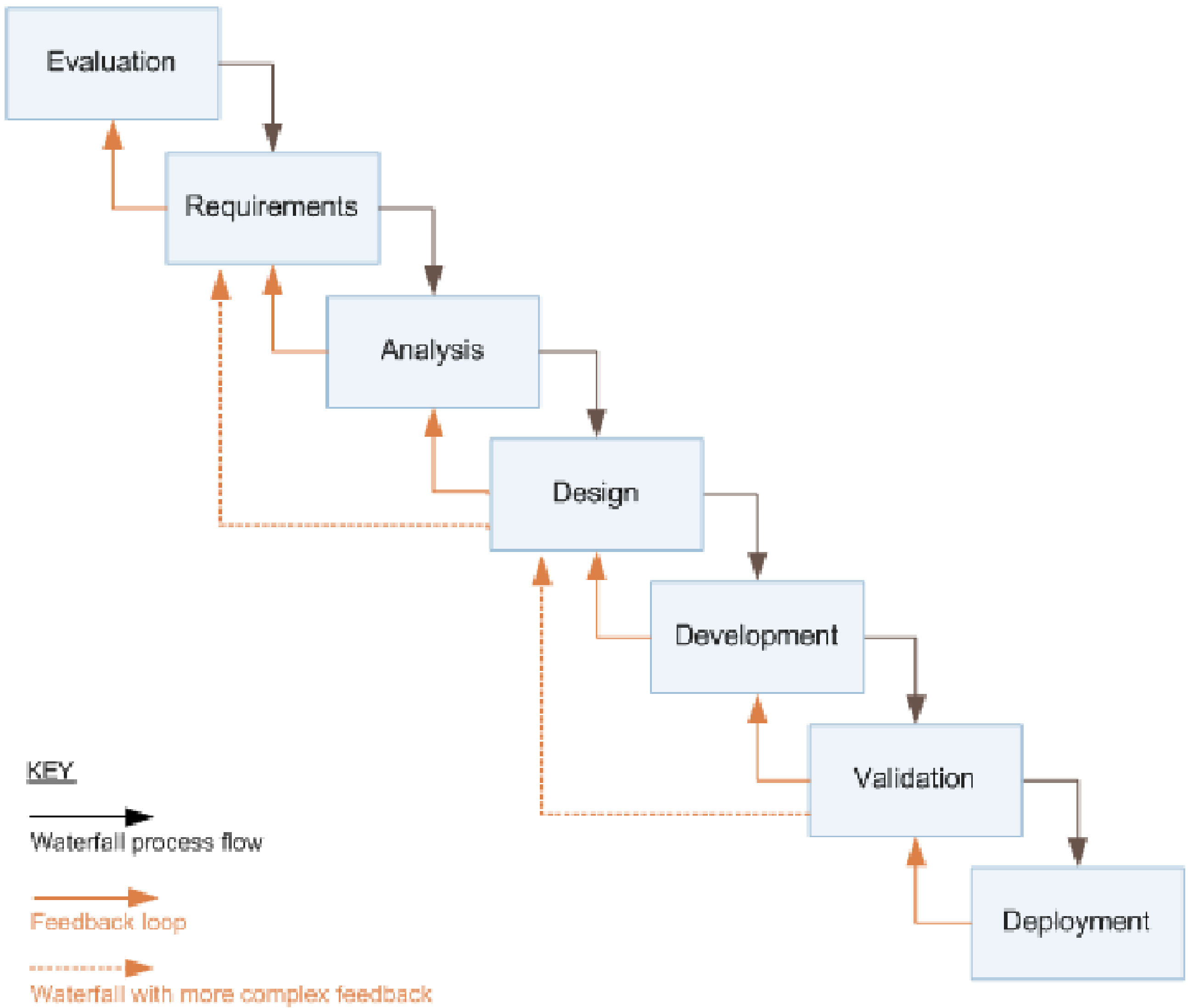

Figure 1. Waterfall methodology with Royce's iterative feedback loops. Reprinted from "Software Development Lifecycle” by N. Ruparelia, 2010, *ACM SIGSOFT Software Engineering Notes, 35*(3), p. 9.

Royce’s iterative structure shown in a Figure 1 is an approach to software development that represents well-structured and robust process of developing a software. The iterative structure of feedback loops creates more flexibility compared to original one-way model which allows to return to the previous steps to revise some of the decisions if they were deemed insufficient at the later steps. Furthermore, this approach provides built-in quality assurance as each step has some sort of validation incorporated into it. However, waterfall methodology has some disadvantages as well. This methodology was initially designed as a rigid sequential flow that lacks flexibility required for developing customer centered software. This approach is the most suitable for a small-scale software or a project that is not expected to shift requirements during development process. Such applications can be back-end or server-side software that provide data to the front end.

2.2 Spiral methodology

Spiral methodology was developed by Boehm in 1986. Ruparelia (2010) states that unlike sequential waterfall methodology the spiral follows iterative approach in software development. The model addresses some of the problems of the waterfall model, where design step might be unnecessary in some situations and the requirement for looking a step ahead to account for risks and software reusability. This methodology is best described by the idiom “start small, think big”. The development process has 4 distinct phases demonstrated in Figure 2. In spiral model each cycle starts by determining the objective. Then the development proceeds to the next stage: identifying and classifying the risks. If development risks are overshadowing performance risks, the development proceeds as a waterfall model. On the other hand, if the performance risk is greater the model can proceed to the next cycle. After that the development and testing of the application conducted. The last stage is to plan the next cycle of the spiral.

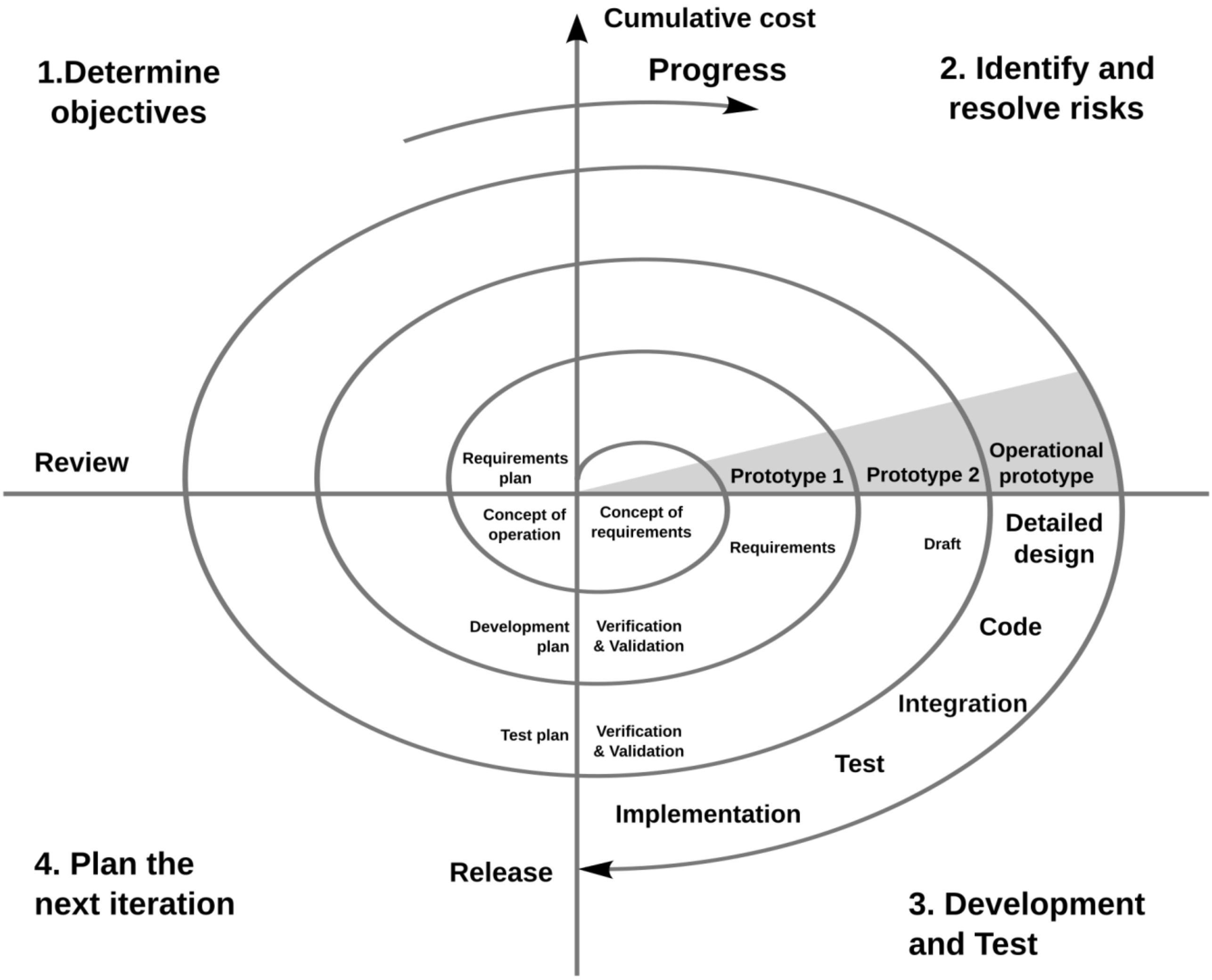

Figure 2. Boehm's spiral life-cycle. Reprinted from ”Spiral Development: Experience, Principles, and Refinements” by B. Boehm et al., 2000, *U.S. Department of Defense.*

The methodology relies heavily on risk management which determines project duration, resources needed as well as management and quality control allocated for the project. Spiral methodology addresses some of the problems of waterfall methodology, namely risks are identified early in the development process and the lack flexibility by allowing to adjust requirements on each cycle of the spiral. However, the model does not address application enhancements and maintenance, and heavily relies on the ability to correctly identify risks in the development process.

2.3 Agile methodology

Agile methodology is one of the most popular models utilized in software development nowadays. According to Ruparelia (2010) “Scope changes, as well as feature creep, are avoided by breaking a project into smaller sub-projects. Development occurs in short intervals and software releases are made to capture small incremental changes.” This approach is often combined with scrum, where development process is broken down to small intervals called sprints.

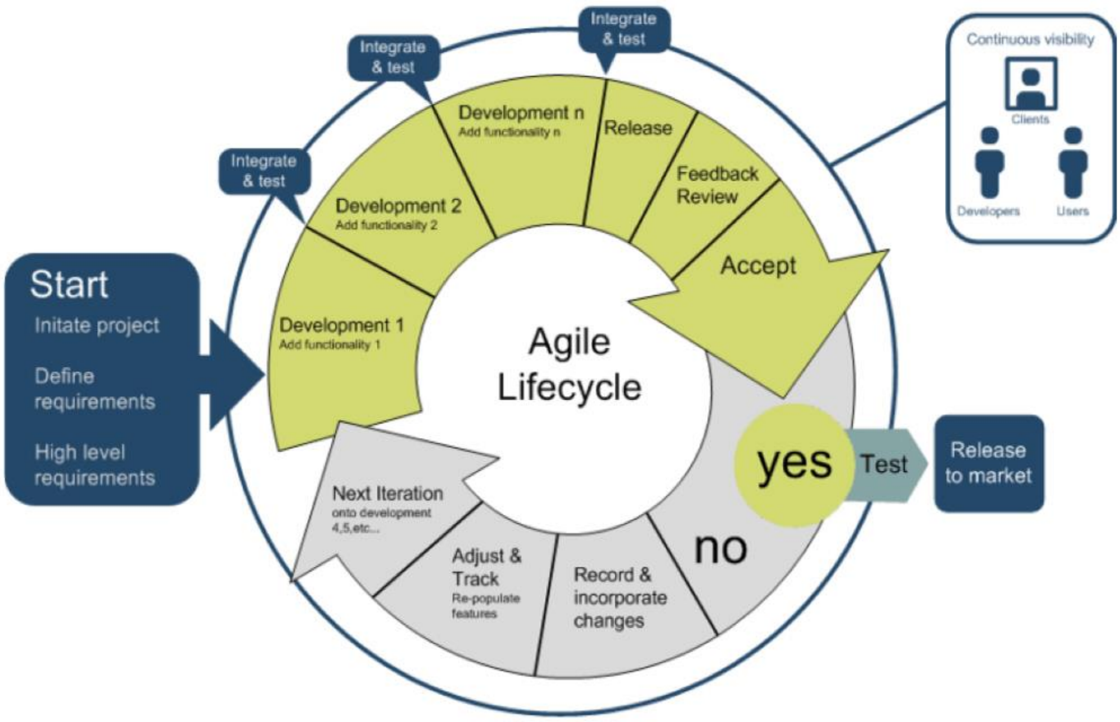

Figure 3. Agile development lifecycle. Reprinted from " Agile Development Process in Software Development" by T. Pham, 2024, *Saigon Technology*

The development proceeds in iterative mode as can be seen from Figure 3. The customer involvement is a big part of the agile development model. Features are developed, integrated and tested. Then based on the customer feedback certain functionalities can be adjusted or changed. This model allows high degree flexibility in the development process. Agile methodology allows to skip the design step and focus on user-driven development. It emphasizes collaboration between development team and end user.

However, agile methodology has some disadvantages. The focus on speed and skipping of designing step might lead to problems down the line as project might encounter compatibility problems. Also, this approach does not enforce a lot of documentation which could make maintenance more difficult and would need to be addressed at some point in the future.

2.4 Methodology comparison

These three methodologies have their advantages and use cases. It is important to choose the approach that suits best for the developed project. The size of the project should be considered as well as complexity, time constraints as budget. Kodmelwar (2022) compares the methodologies based on 17 parameters which can be used to choose the appropriate approach for the needs of each project.

Table 1. Comparison between Waterfall, Spiral & Agile methodologies in software development. Adapted from "A Comparative Study of Software Development Waterfall, Spiral and Agile Methodology," by M. K. Kodmelwar, P. R. Futane, S. D. Pawar, S. A. Lokhande, & S. P. Dhanure, 2022, *Journal of Positive School Psychology*, 6(3), p. 7015. Copyright 2022 by *Journal of Positive School Psychology.*

S. No	Points	Waterfall	Spiral	Agile
1	Method	Sequential method	Evolutionary method	Incremental & iterative
2	Customer	Easy to understand	Tough to understand	Easy to understand
3	Used for Type of project	Small	Large	Large
4	Risk identification	Later stage	Earlier stage	Every stage
5	Flexibility	Difficult to adopt changes	Easy to change requirement	Changes accepted at any stage
6	Risk of use	Higher	Lower	Lower
7	Simplicity	Simple	Complex	Easy
8	Customer involvement	Less	Less	More
9	Deadlines	Large	Large	Short
10	Clarity in requirement	Beginning	Beginning	Requirement dynamic
11	Cost	Fixed	May change with process	Flexible
12	Task	Phases	Iterations	Sprints
13	Focus	Project delivery	Project delivery	Customer satisfaction
14	Testing	After development stage	In each iteration	In each iteration
15	Dependency	Project manager	Project management	Scrum master
16	Team size	Large	Medium	Small
17	Documentation	More	More	Less

Table 1 allows to accurately access advantages and drawbacks of each methodology to perform and informed decision when choosing the appropriate development process. SeatsFlow application is developed by a small team of less than 10 people. As the table management application market already has established companies one of the main goals of the development should be customer satisfaction. It might be difficult to compete with other companies in scope and implemented features as they have higher budgets and have had more time to develop the applications. Thus, it is optimal for this project to focus on creating customer focused service that can address the issues that other companies might have. This requires a flexible development methodology that can enable project to move quickly and adapt to customer feedback. The team requires an easy-to-use approach that allows to rapidly publish new features. In addition, the testing should be conducted often to ensure that updates are immediately available to the customer without any bugs. What is more, the lower documentation requirements can help to make the development process more efficient. However, the proper comprehensive documentation would need to be addressed in the future when the scope of the project and the size of the team expands. In addition, the lower risk of use and dynamic requirement will allow to adapt of the needs of the project swiftly. For these reasons, the agile methodology suits the best for the current project. Sprints can help to manage project in a team environment with a tight schedule and ensure that all the members have a solid understanding of the current state of the project.

To sum up, the agile methodology is chosen to dictate the process of application development. It has advantages that allow to quickly publish new features and quickly adapt to customer feedback and shifting environment. However, there are some methodology specific aspects that affect the development strategy. This approach usually deals with small tasks, so big features will need to be broken down to smaller sub-projects. Also, the customer feedback plays important role in which features should be prioritized and what changes should be implemented.

3 Software types in table management

Before starting the development of the project, it is essential to understand what kind of software is being developed. There are 2 main software solutions that aid restaurant reservation and table management functions: customer relationship management (CRM) systems and table management applications. To determine the scope and the direction of development it is necessary to distinguish between these 2 types. CRM systems usually have a wider range of functions and include marketing, communication, social media integrations and other important features for the business. However, they are not included in the scope of the project as the research focuses on the development of table management applications which focus on internal restaurant management and some of the customer relationship tasks. Therefore, the understanding of both types should be established before starting the development process.

3.1 Customer relationship management (CRM) system

Customer relationship management (CRM) means establishing connections with customers and collection of usable data that can later on be converted into profit for a company (Urbanskienė et al, 2008). Thus, CRM system is a software that helps to facilitate this relationship and to collect, manage and analyze data and convert it to usable information that can be used to enhance business processes. Customer relationship management systems provide a wide range of functions to assist in business. Some of the benefits that they include are lead management, contact management, reporting and analytics, marketing, pipeline management, document management, forecasting, etc. (Gitanjali, 2020). In other words, customer relationship management systems are fairly complex and provide assistance in various aspects of business process. CRM systems enable companies to store information about their customers that can be used to create customer profiles and analyze how they interact with the service. This can lead to better targeting, greater efficiency and increased customer satisfaction. Some CRM applications provide integrations with marketing services, but even without them customer relationship management helps in identifying customer needs and allows to curate marketing materials for specific audiences. Reporting and analytics provide greater insight into operations and enables companies to perform more data driven decisions. What is more, CRM allow companies to improve predictability and plan ahead with data forecasting.

However, CRM systems have some drawbacks as well. According Urbanskienė et al. (2008) implementation of CRM system does not guarantee success and cannot ensure that a company will become customer oriented just by adopting CRM application. Thus, some restaurants and cafes might not want to implement or have a need for a such complex system that would require business strategy adjustments to be successful. What is more, customer relationship management

(CRM) systems despite being useful for customer related tasks might lack in business specific features unique to restaurant management, such as convenient UI and functionality for table management and reservation control.

3.2 Table management application

Table management systems are more lightweight applications that do not provide as much functionality as CRMs. Thus, they are easier to integrate with other software that might already be in use without functionality overlap. Also, such applications are generally cheaper than full CRM systems and do not require as much setup. What is more, being a highly specialized software table management platforms provide a wider range of features and customization in the area of application.

According to Ware (2024) table management platforms enhance operational efficiency in managing dining areas and reservations as well as allow better organization of seating arrangements, reduce wait time for customers and optimize table turnover. Table management applications achieve this effect through comprehensive and visual restaurant layout representation, customer data management and the availability of real-time table occupancy information as well as future reservation statuses. As described by Ware (2024) in the same blog post, most of the table management platforms provide the following benefits to enhance restaurant operational efficiency:

- Efficient seating management
- Real-time table tracking
- Improved customer experience
- Data-driven insights
- Higher turnover rates
- Reduced no-show rates
- Enhanced team coordination
- Integration with POS systems
- Advanced reporting capabilities
- Flexible accessibility
- Customizable layouts
- Automated reservation confirmations

Though the advantages may vary between table management application this list includes the general overview of the benefits they provide. To make a functional table management platform it is important to consider these factors and assess whether the application meets these criteria.

4 Implementation

The implementation process of the thesis focuses on assessing the similar application on the market to determine the essential functionalities for a table management application. This research will provide necessary data to make informed decisions on the development areas and strategies. The information is gathered from the official websites of the companies as well as through personal use of some of the applications listed below. The research will aim to evaluate technical capabilities of the applications and how they compare to SeatsFlow application. However, some of the business aspects of the companies would not be considered. For example, the monetization strategies of these software solutions are not assessed as a part of the research. Most of the applications charge a monthly fee for usage of their services. In addition, some charge addition fee per reservation through their application. What is more, there are application that provide services free of charge with limited functionality. Although, finances and budget affect the development, there are many other factors that affect the choice of the monetization for an application, thus there might not always be a direct correlation between technical aspects of the service and the price. Thus, it is not related to the goal of the project to develop the application's functionality.

After the data is gathered and analyzed, it should be utilized to draw meaningful conclusions about the weaknesses of the current version of the application. The development should focus on these weaknesses, and they should be categorized to prioritize the most essential features. This should enable the beginning of the development process. The new functionalities are designed, and mockups are created to prepare these functionalities to be implemented in practice. The actual coding process is not in the scope of this research for several reasons. Firstly, the programming implementation might be fairly complex for bigger systems and might require input from people with various skillsets such as front-end development with react or back-end development with flask. Secondly, these implementations might be software specific and do not help to answer the broad questions about development of table management applications that this research focuses on. Thus, the thesis is focusing on broader picture of functionality of the software and in not going in depth on the programming aspects.

4.1 Analysis of table management applications

One of the ways to establish areas of development for an application is to analyze and compare current state of the software with competitors in the same market. In the personal survey conducted with restaurant managers the most prevalent applications used for table management were Restoplace, ReMarked and LeClick. These applications will be used to form a basis for analysis of functions table management services usually provided. It is important to understand how these applications generate value for the customer and what features they have that are not present in the developed application.

4.1.1 Restoplace

Restoplace (https://restoplace.net/) is a software application that facilitates management and accommodation of guests in restaurants and cafes. This application provides a wide range of tools for hosting staff to perform their job in an efficient manner. The main features of this application include table reservation book, booking widget, deposits, banquet reservations, SMS notifications, statistics and analytics, guest base, staff management and integration with different services.

Table reservation book enables hosting staff to keep track of table reservations, add, delete and edit reservation requests. Main advantages of the feature are the reminders about upcoming reservations for restaurant employees, table change function that allows to reseat guests, and group booking functionality that allows to reserve more than one table at once for big groups of customers.

Booking widget integration with custom websites, Instagram, Facebook and Vkontakte allows guests to choose and book free tables. The widget provides hall scheme, table photos and information on free and soon to be free tables. This allows customers to make informed decisions about table reservation and make bookings through social network platforms without the need to call restaurants for reservation.

Deposits make it possible for restaurant to set up fees for table reservations. This allows higher end restaurants with booking fees to integrate payment methods into the booking process and create seamless experience for the customers.

Banquet reservations allow customers to book entire halls for big events and weddings or partial hall reservations for corporate events. This feature provides customers with an opportunity to book restaurants for a lot of guests at once without the need to perform table reservations separately.

Reservation notifications via SMS allows to send guests reminders about upcoming reservations. This feature make service more convenient for customers and reduces non-show rate which helps to improve operation efficiency.

Statistics and analytics function help to evaluate the work of employees, analyze attendance by hour, day or month and track booking stats. This tool enables managers to track reservation source, attendance levels, busy hours, new reservation and re-booking rates and more. It provides useful data that can be used for improving efficiency of the service, lowering wait time, avoiding overbooking and allocating staff according to the demand.

Guest base stores information on customers. It helps to catalogue regular guests, track frequency of visits and calculate guest rating. Also, it provides functionality to add tags to customers to track their status. This can improve provided service and increase customer loyalty. Also, it makes possible to prevent people from making reservations by blacklisting them from an establishment.

Access right management for staff allows to limit functionality of an application depending on the permission level and responsibilities. Creator has full control of the application, meanwhile hostess only handles reservation requests and manages seating of guests. This provides more security and access management; at the same time, it simplifies the UI by hiding not needed features from the staff.

Integration options with accounting programs. The service has integration with iiko, r_keeper and poster. These are popular management and automation systems for restaurants and kitchens. Integration with them allow to reduce manual work, increase efficiency, make table management more seamless and improve data collection and tracking.

4.1.2 ReMarked

ReMarked (https://remarked.ru/) is a CRM system focused on management of restaurants and cafes. This application has a wider scope and provides more functionality to customers than regular table management applications. It incorporates the features of Restoplace mentioned previously, such as table reservation book and restaurant scheme, guest database, customer tags, reservation widget and integration with accounting programs, etc. On top of that ReMarked provide WhatsApp notifications and integration with analytics services.

What is more, ReMarked has analytical tools for advertisement tracking. The application provides many customer management tools such as lifetime value (LTV) of a customer and customer journey map (CJM). This functionality is necessary for a CRM system. However, it goes out of the scope of table management applications. Thus, these capabilities will not be considered in the research as they are not related to table management application and are not feasible at the current stage of development.

4.1.3 LeClick

LeClick (https://remarked.ru/booking) is another CRM that is popular among restaurant managers. It is a CRM system, but the company also provides table management application through their partner company SmartReserve. SmartReserve (https://smartreserve.pro/) is a table management application targeted at restaurants, cafes, night clubs and saunas. This application provides customers with similar services to Restoplace. As the application being developed is defined as table management application, it would be compared to SmartReserve instead of the LeClick CRM system. The company provides functionalities for phone reservation management, external website reservation widget, guest data gathering, table management, customer waiting lists and offline functionality for establishment management.

4.1.4 Current state of the SeatsFlow application

To determine development path, it is necessary to access the current state of the application.

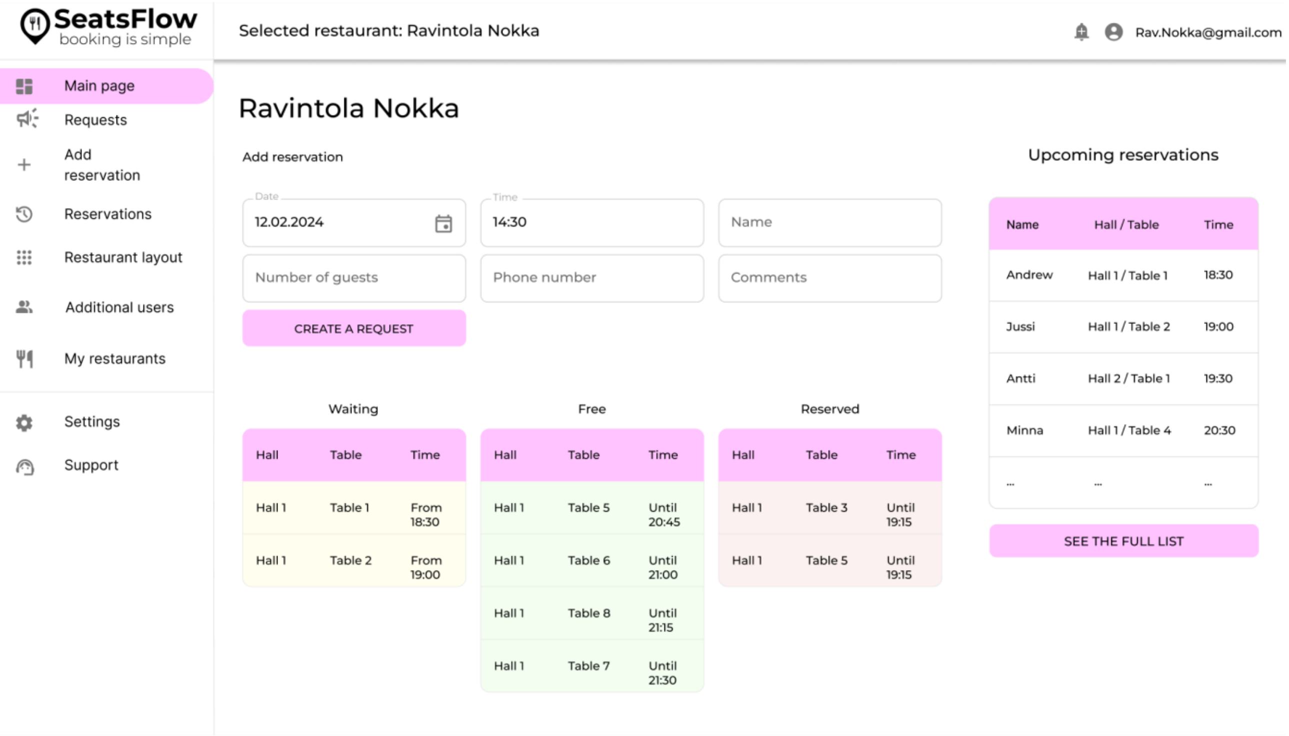

Figure 4. Main page of the SeatsFlow application. Reprinted from SeatsFlow, February 2024.

Figure 4 presents the home page of the application. In the middle is the functionality for creating reservations. It allows for hosting staff to input guest’s name, phone number, number of guests as well as select the specific table to create reservation for. In addition, the hosting staff can add a comment to a reservation in case there are some specific conditions that need to be tracked for the reservation. Below “create reservation” section is a status section. It displays the current status of the tables. On the right side is the list of upcoming reservations. It displays a few closest reservations for the day.

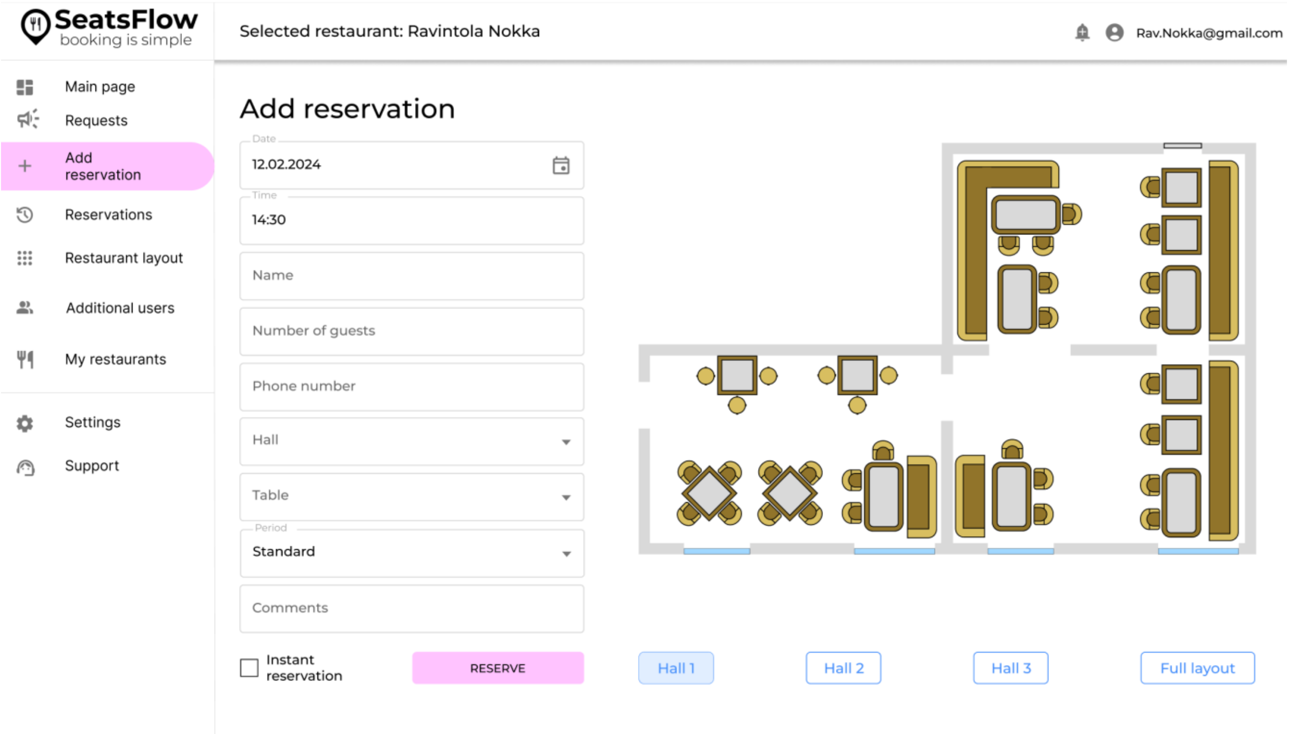

Figure 5. Add reservation screen of the SeatsFlow application. Reprinted from SeatsFlow, February 2024

Add reservation screen allows to create reservations as well as the main page as can be seen from Figure 5. It provides scheme of the establishment to help monitor the tables with a visual display. This function is useful when creating reservations for future dates as status of the table can be easily assessed from the scheme for the selected date. The color of tables changes depending on the status. The grey tables are free, red tables are occupied and yellow color means that the table will be occupied soon. Restaurant layout tab mirrors the scheme of the restaurant that can be seen in Figure 5. It also has a time selection to provide restaurant staff with functionality to check reservations for particular time and date through visual scheme. The usability of the restaurant layout tab is more significant on tablets and mobile devices as it allows to view the scheme on a small screen without abstractions.

On the left side of the application screen is a menu. It contains other functions of the application. The request tab is supposed to have requests for reservations. It is useful for external reservations through restaurant's website. It should contain information for customers reservations that is waiting for approval from restaurant staff. This tab is implemented, so the external reservations do not get added immediately to the booking list to prevent people from disrupting restaurant services through false reservation requests. However, currently this functionality is not in use as the application does not provide widgets to facilitate external reservations through restaurant website. It should not currently be visible for customers on the production version of the application.

Reservations tab contains information on the current, upcoming and past reservations. It contains information about the date and time of the reservation as well as customer name, phone number and optional comments. Through this tab restaurant staff can also cancel and modify reservations as well as mark non-show reservations and extend the time of the reservations.

Additional users tab allows restaurant owners to provide their employees with an access to the application. Another function of this tab is to manage permissions for the employees. Owner has an unrestricted access to the application. Restaurant managers have ability to modify restaurant layout by adding or deleting tables from the scheme. In addition, restaurant management can modify default reservation time and create custom reservation profiles. Hosting staff has access to features related to their job, such as creating and approving reservations as well as modifying reservation times for particular reservations.

My restaurants tab is utilized for adding and management of restaurants. If the client is a restaurant chain or has multiple establishments, from this tab the settings for the restaurants can be changed. This includes modifications to restaurant layout, creation and deletion of halls and tables for a restaurant, modifications to default reservation time and addition of new reservation profiles.

4.1.5 Functionality comparison

After establishing current state of the application, development methodologies and researching competitors on the market, it is essential to define the areas of the application that need further development and implementation. This can be done through comparative analysis of the features that each service provides.

It is worth noting that Table 2 contains a rough overview of the features. The level and quality of implementation may vary from application to application. Even if the software is marked for having certain functionality, the user experience from this feature might not be consistent through all the applications. For example, the POS integration is a broad term describing the connectivity of the application with the point of sale (POS) software. However, different company might partner with different POS service provider. Although, it should be considered that Table 2 generalizes the features into bigger groups, this allows for an easier assessment of the broader landscapes of the applications. Being overly specific might disrupt the process of assessment and lead to subjective or unreliable results. Thus, the general functions of the applications were divided into 12 groups that cover the main features of table management applications.

Table 2. Comparison of table management application functions.

Feature name	Restoplace	ReMarked	LeClick (SmartReserve)	SeatsFlow
Booking widget	Yes	Yes	Yes	No
Deposits	Yes	Yes	No	No
Guest base	Yes	Yes	Yes	No
Online advertising management	No	Yes	No	No
PBX telephone system	No	Yes	No	No
POS integrations	Yes	Yes	Yes	Yes
Restaurant scheme	Yes	Yes	Yes	Yes
SMS notifications	Yes	No	Yes	No
Staff management	Yes	Yes	Yes	Yes
Statistics and analytics	Yes	Yes	Yes	No
Table reservation book	Yes	Yes	Yes	Yes
WhatsApp notifications	No	Yes	Yes	No

CRM systems provide more functionality than table management applications. Thus, some of the features are out of scope of development. As Doan (2024) describes private branch exchange (PBX) telephone systems provide call management, call transfers, customer greetings, reporting & analytics, etc. Despite being useful for managing reservation calls and providing information to the business, such systems require high amounts of integration work. What is more, these services require cloud service providers or hardware infrastructure which lead to additional expenses for development and maintenance. In addition, they have diminishing returns for smaller restaurants and businesses. Thus, they should not be considered for small scale table management applications.

SMS notifications are a useful feature for restaurant booking. It can send reminders to customers and reduce non-show up rate for the business. However, it requires integration with SMS Gateway providers which leads to additional fees. This feature should be considered for medium and large-scale table management solutions. On the other hand, it can be challenging for startups and small-scale table management systems to allocate additional budget on early stages of development. Integration of this functionality depends on company revenue and investments. In case of startup companies with no external investments it might not be feasible to increase costs of the project. Thus, the need for additional operational costs decreases the priority of implementing SMS notifications feature.

Online advertising management is a feature implemented by Remarked requiring integration with customer’s websites and online services. This functionality is relevant to CRM systems and provides more ways for customers to gather information. However, for internal table management systems it is not a necessity. As can be seen from the Table 2 only one of the competitors provide this functionality. Thus, it is not essential for table management systems and is out of scope for the developed application.

Reservation deposit system allows restaurants to charge reservation fees through external reservation widgets on restaurant websites and applications. It requires integration with banking systems as well as infrastructure for restaurants customers to interact with table management application. The developed application lacks functionality for restaurant website integration. Thus, before considering implementing reservation deposit system, development of functionality for external widgets is required.

Booking widget is one of the features that can be seen in all other analyzed applications. This indicates that implementation of this feature should be considered as one of the first priorities to be able to compete on the market. However, market for the application should be considered before deciding whether this feature should be prioritized. It is difficult to compete in a market with already established competitors. Thus, the primary customer group for the developed application is small restaurants that do not use any table management system. Such customers can be small and family-owned restaurants which might not have personal websites. Thus, development of a widget that can be integrated into restaurant website does not always provide customers with more value from the service. In addition, the application is focused on internal use for restaurants and cafes. To avoid scope creep, it is essential to define boundaries for the project. The booking widget adds another layer of direct customers interaction with the application. This addition will require many human-hours for backend development to implement server-side data handling as well as frontend development to implement widget interface. What is more, this time could be spent developing other areas of the application. Thus, booking widget, despite being available in other table management applications, is not included in the scope of the current development cycle.

Guest database and statistics & analytics systems are implemented by all the analyzed competitors. Thus, these can be considered essential features for table management software. The development of the application should be focused on implementation of guest base and analytics tools to provide value to the customers and to be able to compete in the table management application business. These features can be classified as a subset of business analytics tools which can provide restaurants with actionable insights for their operations and reinforce data driven decision making. International Institute for Management Development (2024) describes benefits of the business analytics tools through increased operational efficiency, insights into customer behavior, predict impact of company initiatives, facilitate more informed business decisions, evaluate performance through KPIs and leverage patterns & trends for competitive advantage. Thus, these features provide much value to the customers.

To sum up, analytical tools are essential for businesses. Thus, the development of the application should focus on implementation of guest databases and statistics & analytics systems the most. Guest databases provide functionality to collect customer data. This information then can be used to create more accurate and insightful analytics data. Thus, the guest database is one of the features that the development team should focus on first.

4.2 Direction of development

As defined in chapter 2.4 the project follows agile development methodology. This facilitates frequent testing and feedback from the customers. In the personal interview with one of the restaurant managers after testing the application they underlined some features that can be essential for efficient use of the application. These features include phone input mask and ability to collect data on a customer.

4.2.1 Phone input mask

Reservation area on the main page of the application is the most used part of the service. Customer reservations are accepted through phone calls. This necessitates reservation inputs with restaurant customer’s information to be as efficient and convenient as possible. One of the complaints was the lack of phone formatting when inputting it to the reservation form. The need to input country code for every table reservation makes the process less efficient and the lack of formatting make it difficult for hosting staff to identify the numbers from the first glance. Microsoft defines input mask as a string of characters that indicates the format of valid input values. Thus, the formatting instructions should be implemented for the phone input.

```
export const formatPhoneNumber = (value: string | IGuest) => {
  if (!value) return "+7 ";
  if (typeof value === "object") {
    value = value.phone;
  }
  let cleaned = value.replace(/\D/g, "");

  if (!cleaned.startsWith("7") && !cleaned.startsWith("8")) {
    cleaned = "7" + cleaned;
  }

  let formatted = cleaned.startsWith("8") ? "8 " : "+7 ";

  if (cleaned.length > 1) {
    formatted += cleaned.slice(1, 4);
  }
  if (cleaned.length >= 5) {
    formatted += ` ${cleaned.slice(4, 7)}`;
  }
  if (cleaned.length >= 8) {
    formatted += `-${cleaned.slice(7, 9)}`;
  }
  if (cleaned.length >= 10) {
    formatted += `-${cleaned.slice(9, 11)}`;
  }

  return formatted;
};
```

Figure 6. Phone formatting function for the booking form of SeatsFlow applicatioin.

The Figure 6 contains an implementation of the function that can be used for masking phone number in the phone input field of the booking form. As the application provides services in only one country, the country code can be persistent part of phone input. The phone input field should only accept numerical characters. The length of the input field should be limited to 11 numbers. The common ways of input formatting can be “+7 XXX XXX-XX-XX” or “+7 (XXX) XXX-XX-XX”. However, the second option add 2 formatting characters at a time shifting the input cursor 2 spaces at once.

This approach might be confusing in cases where restaurant staff needs to correct a single digit in the input. So, the first input formatting opting is better for this use case.

In addition, the input should be able to handle copy & paste functionality. First the input should be sanitized by extracting only numerical characters from the string. Then it should check for input length. If it is 11 characters, it should delete the first number and accept the rest as country code is already included in the formatting. Otherwise, the numerical characters should be added to the input field without changes and the field formatting should be applied to the number.

4.2.2 Guest base

In the center of the restaurant business are the guests. Thus, tracking their information provides restaurants with valuable data and can improve service for restaurant's customers. The guest base can be considered one of the most important features of the table management application. As the service primarily works with phone reservations the phone number should be used as an identifier for guest base entries. If customer calls from the same number but introduces themselves with different name or if hosting staff records customer name in a different way, application should have functionality to store multiple names for a single customer. The name used on the creation of the guest base entry should be used as a primary name. The restaurant staff should have an ability to change the primary name of the entry. The mockup of this feature is demonstrated in the Figure 7.

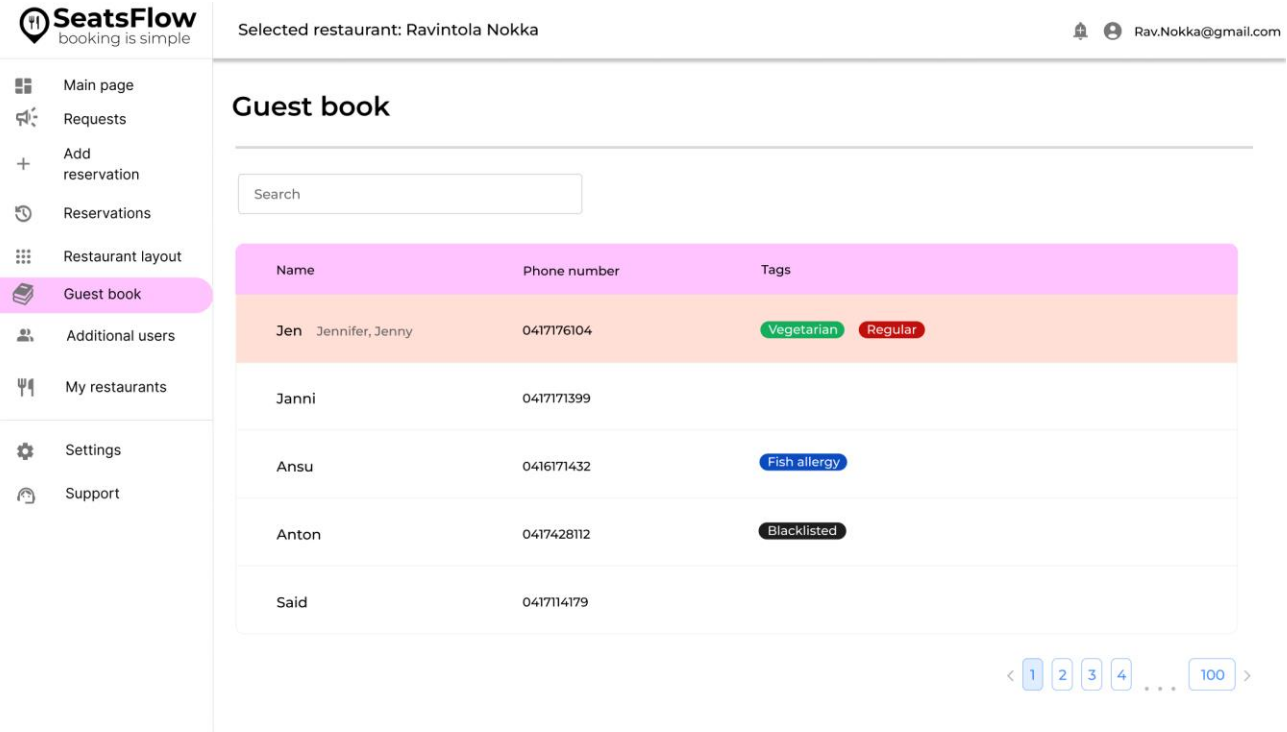

Figure 7. Mockup of the interface of the guest book for the SeatsFlow application.

In the guest base it should be possible to select a single entry. The entry should contain primary name as well as possible additional names and phone number. In the entry it should be possible to add a VIP status to the guest. In addition, the entry should contain the total number of visits. There should be possibility to assign favorite table for the customer. Also, the entry should have a field for a comment where any additional information on the customer can be written. The mockup for individual guest base entry can be seen in the Figure 8.

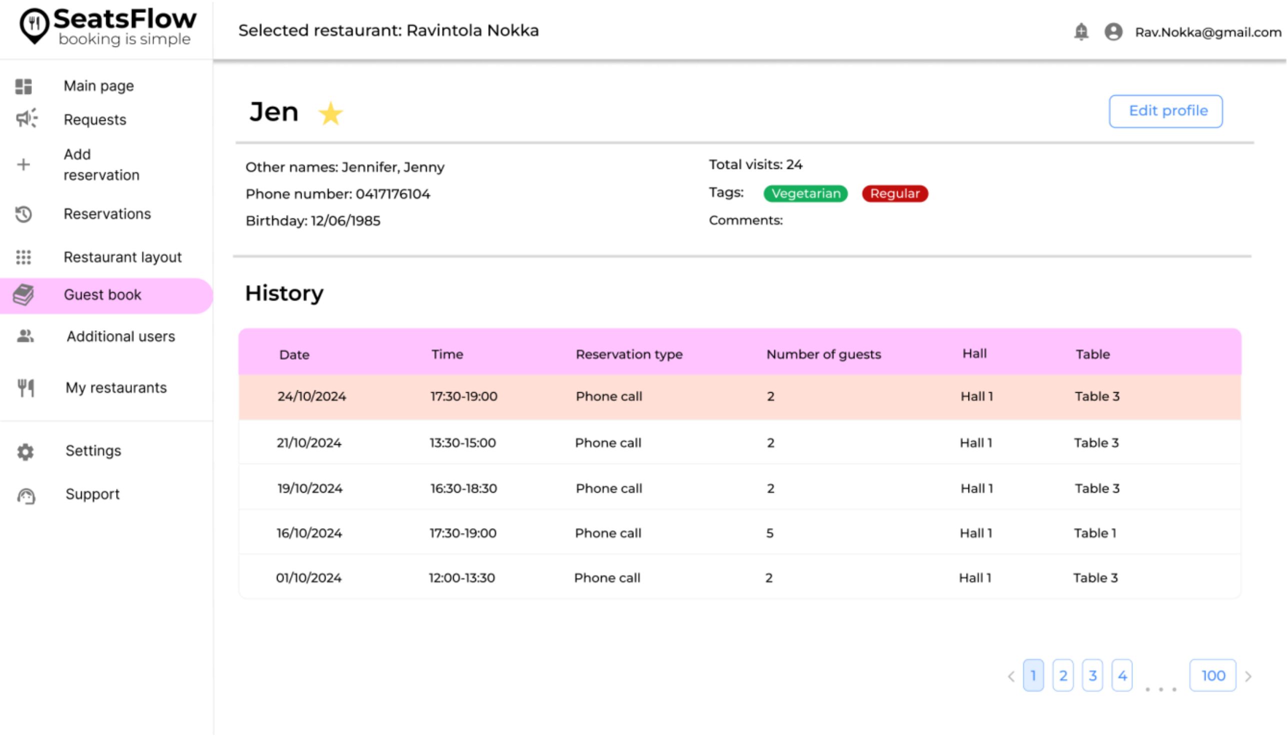

Figure 8. Interface for individual guest book entry for the SeatsFlow application.

What is more, the guest base should support tag system. It should contain a list of default tags like vegetarian, regular customer, blacklisted, etc. Restaurant staff should be able to assign tags from the list of default and be able to create custom tags. In addition, the individual guest base entry should have a history of visits for every customer. The history should display the date and time of reservation, number of guests per reservation and reserved table.

4.2.3 Statistics & analytics tools

Statistics and analytics tools provide application customers value through information gathering and transformation of the information into usable data. The implementation of the guest base will allow collection of the data on the customer which can be analyzed and used to get more insight into restaurant operations. The data should be displayed using interactive graphs to create visual data that can be easily comprehended. The mockup of the analytics page could be seen on Figure 9. Though this demonstration is very limiting as it doesn't display any interactive elements of the charts.

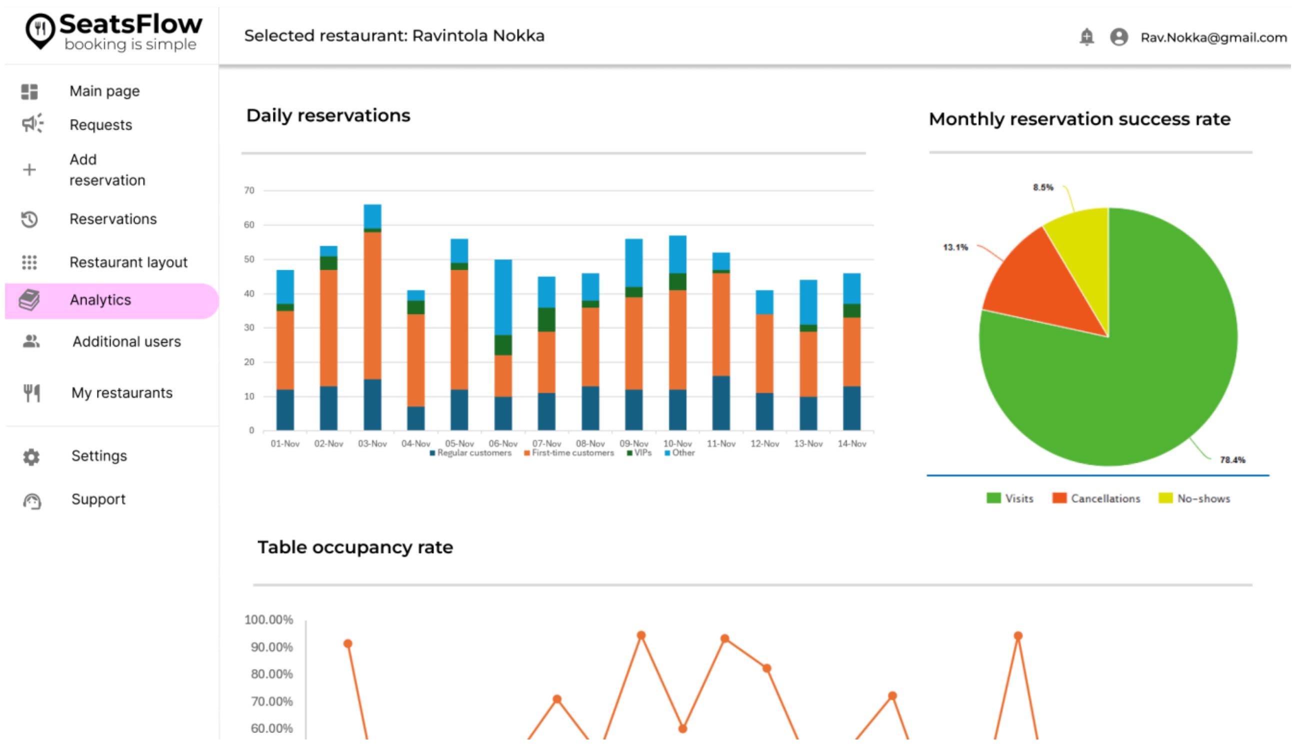

Figure 9. Mockup of the analytics page for the SeatsFlow application.

The application should track the number of monthly and daily reservations. The reservations should be divided by the status: arrived, canceled, non-show. This data should be used to create a graph. For temporal data a line chart, spline line chart or stacked area chart should be used. To visualize monthly visitor the spline line chart is the best choice as it smooths the curved between data points better demonstrating dynamic between them rather than focusing on the individual points. Daily data can be visualized using bar chart to focus on individual data points by their type while still demonstrating data trends over the period of time. In addition, the monthly and daily data can be divided by the customer type to visualize the number of reservations between regular customers, VIPs and first-time customers. These charts can provide information on the average number of visitors or reservations in the restaurant as well as what types of guests are served by the restaurant.

A pie chart can be utilized to visualize what part of reservations resulted in a customer visit against cancelations and non-shows. This data can provide a valuable insight into what percentage of the reservations turn into guests.

The other type of data that can be plotted over time is an average wait time per customer. This can be useful for establishments with high demand. The data can help to identify bottlenecks and rush hours in the restaurants. It can be applied for planning employee's shift to allocate more staff for busy hours.

What is more, the analytics can track average reservation time. By default, the application allocated 90 minutes for each reservation. However, this time may vary depending on the restaurant, the time of day and the day of the week. So, additional reservation profiles can be created to facilitate service based on the time and occupation of the restaurant.

In addition, the occupation rate per table and occupation rate per day/week can be displayed with a spline line chart. It serves similar purpose to wait time chart and provides the restaurants with information on the demand and customer rate. The data can be used to improve efficiency of the restaurant service and to provide customers with information on the best visiting time.

These data charts should provide restaurants with actionable information to perform informed decisions. The functionality should be tested within restaurant environments to acquire feedback on the usability of the data. Depending on the information received the types of graphs can be adjusted for better visualization and other types can be added if restaurant management require additional data metrics for tracking.

5 Results

As the result of the project the assessment of the table management applications has been conducted. The software development methodologies were evaluated and agile was determined as a suitable methodology for a small development team with the focus on the customer relationship. The comparison between the competing applications has been conducted and the common features for table management software were identified. This data gathering helped to conclude which are the essential functionalities that every table management application should have. By analyzing these results, the areas that need to be prioritized during development were identified. Such areas include the guest base and statistic & data analyzing tools. What is more, customer feedback helped to identify the essential feature of phone input masking for table management applications. This led to the development of requirements for phone formatting which includes automatic country code input as well as separation of characters for better readability. What is more, the need for phone input to be able to handle copy & paste functionality was established. In addition, the requirements for guest base were developed. Which should include the customer's phone as identifier, customer name with functionality to store multiple names for a single customer and support for customer tags. Furthermore, individual guest cards mockups were developed to demonstrate how the interface should look like and what functionalities should be displayed. After that, statistics and analytics functionality was developed. The research described possible data visualization methods that should be implemented. In addition, the mockup of the interface for the analytics page was developed.

In the process of the research many limitations of the project were discovered. It established that although some features would be beneficial for the application and provide value to the customers, it is not always possible to implement despite their advantages. Any project has time, financial and human resources constraints which makes the process of prioritizing the most important features indispensable for software development. In addition, the research identified further areas of development which did not fit in the current development cycle. One of the most notable was external widget for restaurant websites. This feature should be prioritized next as it provides a lot of value to the restaurants and can make reservation experience much easier for their customers. What is more, there are a lot of features that can be build on top of the restaurant websites widgets, such as deposit system. This system would be able to make it possible for restaurants charge for reservations and could help restaurants to build online monetization further as a banking system would be already integrated in their service. Even though the project did not explore the programming implementation of the features, the research established a landscape for table management applications and functionalities that such applications should have.

6 Discussion

6.1 Reliability

The research relies on theoretical basis of the software development methodologies and table management application service requirements. The data analyzed is based on feedback from the customers of the application that performed tests in the real working environment. In addition, data used for determining table management application features and how the developed application can be improved is based on the publicly available information on the competitors in the market of restaurant management software. The research relies on data acquired through these methods which was analyzed to draw conclusions and create a development plan for the table management application

6.2 Research results

The software development methodologies were analyzed and compared. This allowed to conclude that the agile methodology should be adopted for the project facilitate high level of cooperation with customers and frequent feedback from testing. Thus, the development process should be adjusted according to customer needs. This strategy proved to be the most suitable for the small team working on the project in an environment with already established competitors. The agile methodology provides many benefits for the project through ease of use and frequent customer feedback.

The table management application functionality centered around increasing efficiency of the restaurant service. Such results are provided by improving reservation creation process. This is achieved through integration with POS systems, visual design and streamlining of the table management process. In addition, table management applications provide value to the customer through data collection and analysis tools. To facilitate efficient restaurant work, table management application should provide actionable data and insights into operation process. The list of common table management functionality might include booking widgets, deposit systems, guest databases, POS integration, customer reservation notifications, staff management, statistics & analytics, table reservation book, and restaurant schema.

The ways that the developed application could be improved was determined through comparison to competitors on the market. The application lacks external widget functionality, phone notifications as well as guest base and statistic and analytic tools. As demonstrated in the research most of the other applications in the table management field provide such capabilities to their customers. Thus, application can provide more value to the customer through implementation of these features. However, the possibilities for improvement are not limited by these functionalities. The table management applications are complex systems, and they can be developed to include online marketing monitoring tools and AI models for data analysis and predictions.

The development process should focus one the implementation of the guest base and statistics & analytics tools as they have the most impact on the restaurant management process. What is more, their implementation does not create additional maintenance cost and does not increase the scope of the project. Thus, the ratio between development time and cost against value provided to the customers of the service is the highest for this functionality. However, these are only areas that are most relevant to the development process at the current state of the application. The research describes other functionalities that could be added in the next development cycles.

6.3 Conclusion

The development of software applications requires analysis of the current state of the project. The proposed solutions and development focus areas can be expanded after the current state of the project finishes. As underlined in chapter 5.2, table management applications are complex systems that incorporate a wide range of features. Thus, the service can be analyzed further to determine what other improvements are required to the application. What is more, the research focuses on the functional development of the application features. However, there are other areas that might require further improvement. These areas could include accessibility through UI/UX design and optimization of the current implementation to improve loading times and various screen resolutions.

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